Week 2 Practice Problem Solutions

5. Bank Transaction Reference Generator & Validator

Normalize a reference, validate its 14-character structure, then format the date and sequence.

Java Solution

```
public class Main {
    static String normalizeReference(String raw) {
        String reference = raw.trim();
        if (reference.length() < 3) {
            return reference.toUpperCase();
        }
        return reference.substring(0, 3).toUpperCase()
                + reference.substring(3);
    }

    static String validateAndFormat(String reference) {
        if (reference.length() != 14) {
            return "Invalid: reference must be exactly 14 characters";
        }
        for (int i = 0; i < 3; i++) {
            if (!Character.isLetter(reference.charAt(i))) {
                return "Invalid: bank code must be 3 letters";
            }
        }
        for (int i = 3; i < 14; i++) {
            if (!Character.isDigit(reference.charAt(i))) {
                return "Invalid: reference body must contain only digits";
            }
        }

        String date = reference.substring(3, 9);
        StringBuilder output = new StringBuilder();
        output.append("[").append(reference.substring(0, 3))
              .append("] DATE: ").append(date, 0, 2).append("/")
              .append(date, 2, 4).append("/").append(date, 4, 6)
              .append(" | SEQ: ").append(reference.substring(9));
        return output.toString();
    }

    public static void main(String[] args) {
        String valid = normalizeReference(" hdf03022600042 ");
        System.out.println(validateAndFormat(valid));
        System.out.println(validateAndFormat(
                normalizeReference("12F03022600042")));
    }
}
```

Sample Output

```
[HDF] DATE: 03/02/26 | SEQ: 00042
Invalid: bank code must be 3 letters
```

Notes

The date is rearranged from ddMMyy to dd/MM/yy; the leading zeroes in the five-digit sequence are preserved because the reference is handled as a String.